Exercise Sheet 3

1. Let $\mathcal{H}$ be a Hilbertspace and let S be a closed subspace of $\mathcal{H}$. Denote by $P_S(x)$ the projection of x onto S. Then
   (a) $P_S(\alpha x + \beta y) = \alpha P_S(x) + \beta P_S(y) \quad \forall x, y \in \mathcal{H},\ \alpha, \beta \in \mathbb{R}$.
   (b) $\|x\|^2 = \|P_S(x)\|^2 + \|(1 - P_S)(x)\|^2 \quad \forall x \in \mathcal{H}$.
   (c) $P_S(x_n) \to P_S(x)$ if $x_n \to x$.
   (d) $x \in S$ if and only if $P_S(x) = x$.

2. Assume that $e_1, \ldots, e_m$ are *orthonormal* in some Hilbert space $\mathcal{H}$, i.e.
$$\langle e_i, e_j \rangle = \begin{cases} 0 & \text{if} \quad i \neq j; \\ 1 & \text{else.} \end{cases}$$
   Let $x \in \mathcal{H}$ be arbitrary.
   (a) Show that for $S = \text{span}\{e_1, \ldots, e_m\}$ we have $P_S(x) = \sum_{i=1}^m \langle x, e_i \rangle e_i$.
   (b) Show that $\|P_S(x)\|^2 = \sum_{j=0}^m \langle x, e_i \rangle^2$.
   (c) Conclude from Exercise 1 that $\|x\|^2 \geq \sum_{j=0}^m \langle x, e_i \rangle^2$.

3. Suppose that (Z_n) is a white noise sequence $Z_n \sim \text{WN}(1)$ and let X_n be a time series such that
$$\text{Cov}(X_n, Z_{n-i}) = \begin{cases} \theta_i \text{ for } i \in 1, \ldots, q; \\ 0 \text{ else.} \end{cases}$$
   Let $S = \text{span}\{Z_n, \ldots, Z_{n-m+1}\}$. Apply the result in Exercise 2 to show that:
   (a) For any $m \geq q$ we have $P_S(X_n) = \sum_{i=1}^q \theta_i Z_{n-i}$.
   (b) For any $m \geq q$ we have $\text{Var}(P_S(X_n)) = \sum_{j=0}^q \theta_i^2$.

4. Let $\{X_t\}$ be an AR(2) process:
$$X_t = 0.5X_{t-1} - 0.3X_{t-2} + Z_t.$$
   Provide an explicit formula for the best linear forecast for X_{t+1} based on a sample $X_1, \ldots, X_t$ using (a) 1 lag and (b) $k \geq 2$ lags.

   Hint for (a): Compute the lag-1 correlation of this process by computing $\text{Cov}(X_t, X_{t-1})$ using both sides of the model equation.

5. Let $\{X_t\}$ be an MA(2) process:
$$X_t = 0.5Z_{t-1} - 0.3Z_{t-2} + Z_t.$$

Provide an explicit formula for the best linear forecast for X_{t+1} based on a sample $X_1, \ldots, X_t$ using (a) 1 lag and (b) 2 lags.

6. Consider a linear process

$$X_n = \sum_{k \geq 1} \psi_k Z_{n-k}.$$

Let $\gamma(h)$ the covariance function of the process. Show that $\gamma(h) \to 0$ for $h \to \infty$. Why is this condition important?